Unit 2 - Neural Networks

Machine Learning | Unit 2

Introduction to Neural Networks

Artificial Neural Networks (ANN) are computing systems inspired by biological neural networks in the brain. They consist of interconnected nodes (neurons) organized in layers that process information using connectionist approaches to computation.

Structure of a Neural Network

Input Layer

Receives raw input data. Number of neurons equals number of input features.

Hidden Layers

Intermediate layers that transform input into increasingly abstract representations. More layers = deeper network.

Output Layer

Produces final predictions. Number of neurons depends on the task (1 for regression, N for N-class classification).

Activation Functions

- Sigmoid - outputs between 0 and 1, used in output layer for binary classification
- ReLU (Rectified Linear Unit) - max(0,x), most popular in hidden layers
- Tanh - outputs between -1 and 1, zero-centered
- Softmax - converts outputs to probabilities, used for multi-class classification
- Leaky ReLU - variant of ReLU that allows small negative values

Backpropagation

Backpropagation is the algorithm used to train neural networks. It calculates the gradient of the loss function with respect to each weight by the chain rule, propagating error backwards through the network. The weights are then updated using gradient descent to minimize the loss.

Gradient Descent Variants

Batch Gradient Descent

Uses entire dataset to compute gradient. Stable but slow for large datasets.

Stochastic Gradient Descent

Uses one sample at a time. Fast but noisy updates.

Mini-batch Gradient Descent

Uses small batches of data. Best of both worlds, most commonly used.

Common Problems in Training

- Vanishing Gradient - gradients become too small in deep networks
- Exploding Gradient - gradients grow too large, use gradient clipping
- Overfitting - use dropout, regularization, or early stopping
- Local Minima - use momentum or adaptive learning rates